



higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

**ELECTROTECHNOLOGY
NQF LEVEL 2**

(12040212)

**19 February 2025 (Y-paper)
13:00–16:00**

Calculators may be used.

This question paper consists of 7 pages and 1 formula sheet.

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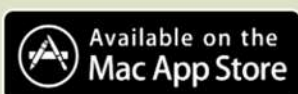
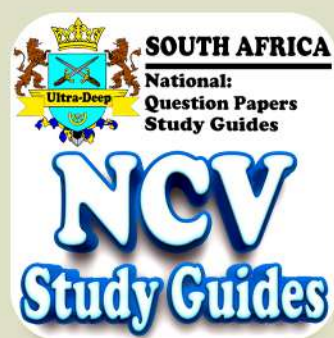
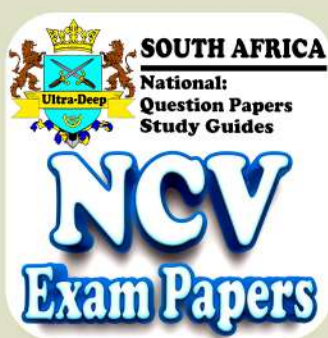


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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA
NATIONAL CERTIFICATE (VOCATIONAL)
ELECTROTECHNOLOGY
NQF LEVEL 2
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
 2. Read all the questions carefully.
 3. Number the answers according to the numbering system used in this question paper.
 4. Start each question on a new page.
 5. Use only a black or blue pen.
 6. Write neatly and legibly.
-

12)

-3-

QUESTION 1

1.1 Indicate whether the following statements are TRUE or FALSE by writing only 'True' or 'False' next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 An inductor stores energy in a magnetic field when electric current flows through it.

1.1.2 A diode allows current to flow in both directions.

1.1.3 A capacitor stores electrical energy in a magnetic field.

1.1.4 Direct current (DC) is an electric current that reverses its direction periodically.

1.1.5 Semi-conductors have electrical conductivity between that of a conductor and an isolator.

(5 × 1) (5)

1.2 An RLC circuit consists of a $12\ \Omega$ resistor, an inductor of $0,2\ \text{H}$, and a capacitor of $50\ \mu\text{F}$ all connected in series. The circuit is connected across a $230\ \text{V}$, $50\ \text{Hz}$ supply. Calculate the following:

1.2.1 Inductive reactance

1.2.2 Capacitive reactance

1.2.3 The impedance of the circuit

1.2.4 The total current

(4 × 2) (8)

1.3 Analyse the circuit in FIGURE 1 and answer the questions that follow.

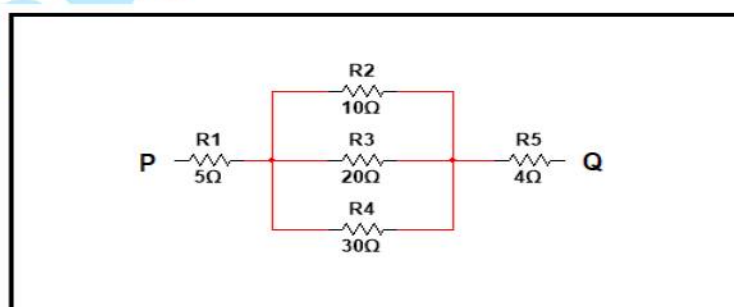


FIGURE 1

1.3.1 Calculate the total resistance between point P and point Q.

(4)

1.3.2 Calculate the voltage if a total current of $12\ \text{A}$ flows through the circuit.

(2)

1.4 State Joule's law.



(1)
[20]

12)

-4-

QUESTION 2

2.1 Draw the IEC symbols for the following:

2.1.1 AC motor

2.1.2 Two-way switch

2.1.3 Coil

2.1.4 Push button switch

(4 × 1)

(4)

2.2 Draw a neat and fully labelled circuit diagram to show how two socket outlets are connected to the same circuit breaker.

(6)

2.3 State the recommended conductor (wire) size used in a domestic installation for the following sub-circuits:

2.3.1 Geyser

2.3.2 Stove

2.3.3 Distribution board wiring

2.3.4 Socket outlet

2.3.5 Bell circuit

(5 × 1)

(5)

2.4 It is essential to understand the difference between various types of conduits used in electrical wiring.

Give THREE advantages and disadvantages of metal conduit.

(6 × 1)

(6)

2.5 Define the following:

2.5.1 Fault current

2.5.2 Earthing

(2 × 2)

(4)

[25]

12)

-5-

QUESTION 3

- 3.1 Identify the components shown in FIGURE 2 by writing only the answers next to the letter (A–F) in the ANSWER BOOK.

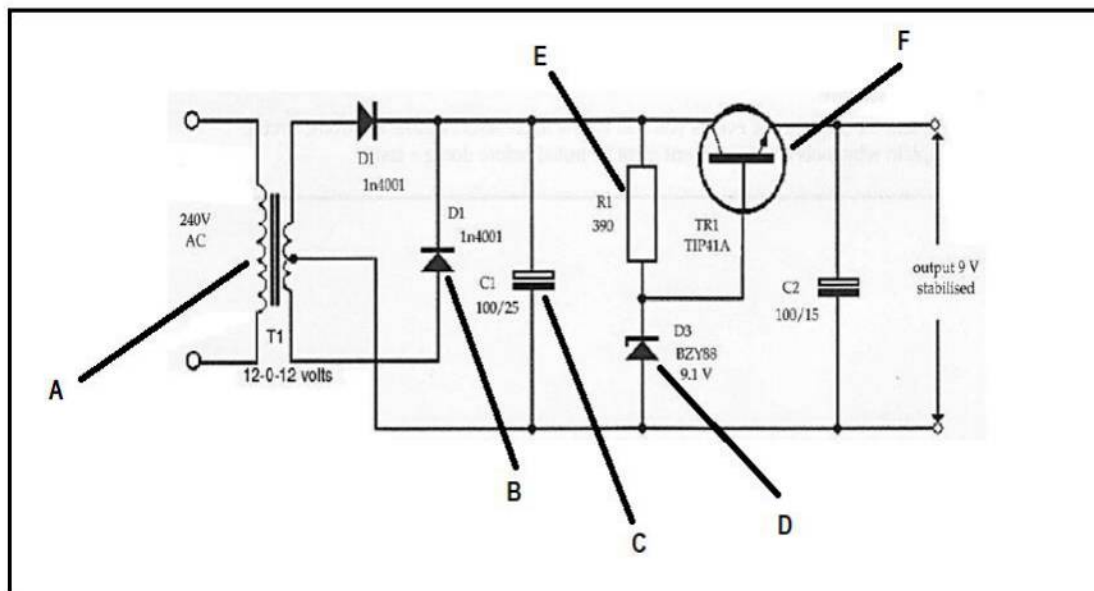


FIGURE 2

(6)

- 3.2 Explain how one would test a transistor using a digital multi meter. (4)
- 3.3 With the aid of a circuit diagram, show how one would connect the following instruments in a circuit: ammeter, voltmeter, and ohmmeter.

Hint: The circuit diagram must consist of a DC supply, two resistors in series, and an open switch. (6)

- 3.4 When planning, it is important to tackle any job in the right order of events.

What are the FOUR questions one should ask oneself when planning a job? (4)

- 3.5 A basic toolkit that contains the basic hand tools needed to do any work should be available in all electronic workshops.

List FIVE tools needed to build an electronic circuit on a Veroboard.

(5)

[25]



12)

-6-

QUESTION 4

- 4.1 Choose a description from COLUMN B that matches a term in COLUMN A. Write only the letter (A–F) next to the question number (4.1.1–4.1.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
4.1.1	High-resistance joint	A	A troubleshooting technique that involves using sight to identify visible signs of faults or damage
4.1.2	Continuity testing	B	A fault condition where connections have higher than normal resistance, causing voltage drops and potential overheating
4.1.3	Corrective maintenance	C	A process of tracing and correcting faults in machinery and technical equipment
4.1.4	Visual inspection	D	Activities performed routinely to extend the life or increase the performance of the equipment
4.1.5	Preventive maintenance	E	A method to test the continuous electrical path in a circuit to ensure that there are no breaks
	★	F	These are activities done because of partial or total equipment failure

(5 × 1)

(5)

- 4.2 Complete the following sentences by writing only the missing word or words next to the question number (4.2.1–4.2.5) in the ANSWER BOOK.

- 4.2.1 Plastic does not conduct electricity because it has ... electrical resistance.
- 4.2.2 Metals are good conductors because they have ... resistance to the flow of electricity.
- 4.2.3 A short circuit provides an ... path for current to flow to earth. ★
- 4.2.4 An open circuit is defined as any ... in the current path of a series circuit that will cause the entire circuit to become dysfunctional.
- 4.2.5 Soldering provides electrical ... between the device and the board.

(5 × 1)

(5)

12)

-7-

4.3

Refer to FIGURE 3 and answer the questions that follow.

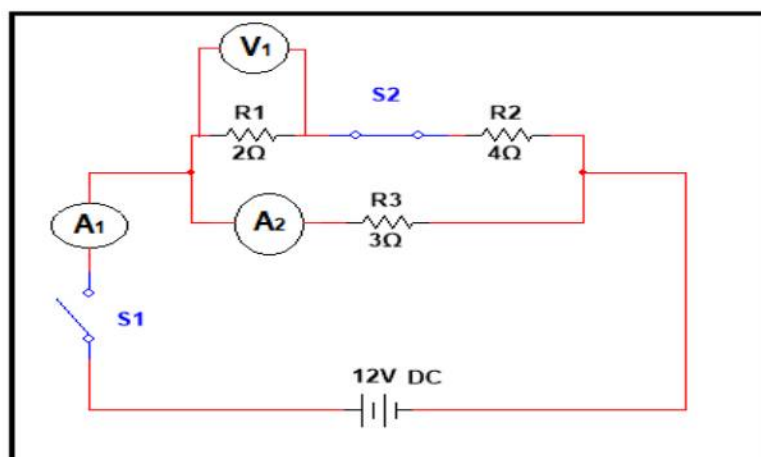


FIGURE 3

- 4.3.1 When switch S1 is closed, what will the reading on meter A₁ be? (4)
- 4.3.2 What will the reading on meter A₂ be? (2)
- 4.3.3 What will the reading on ammeter A₁ be, if switch S1 is closed but switch S2 is opened? (2)
- 4.3.4 What could the possible cause for the following readings be? (2)
- V₁ = 0 V
 - A₂ = 0 A

[20]

QUESTION 5

An electrical contract is made up of activities A to E.

- Activity A takes 2 weeks, starting in week 1.
- Activity B takes 3 weeks, ending in week 6.
- Activity C takes 3 weeks, starting in week 3.
- Activity D takes 2 weeks, starting in week 6.
- Activity E takes 4 weeks.

Note that Activity B does not prevent any other activity from taking place, but Activity E can only start after Activity C has been completed.

- 5.1 Draw a bar chart to display the activities of the contract. (8)
- 5.2 Due to unforeseen circumstances, Activity C takes 5 weeks to complete. When will Activity E be completed? (2)

[10]

TOTAL: 100

12)

ULA SHEET

ELECTROTECHNOLOGY LEVEL 2

1. $P = V I \cos \phi$

2. $Q = V I \sin \phi$

3. $S = V I$

4. $V = I R$

5. $P = V I$

6. $X_L = 2\pi f L$

7. $X_C = \frac{1}{2\pi f C}$

8. $Z = \sqrt{R^2 + (X_L - X_C)^2}$

9. $V_T = \sqrt{V_R^2 + (V_L - V_C)^2}$

10. $V_1 = \left(\frac{R_1}{R_t} \right) V_T$

11. $V_L = I X_L$

12. $I_T = \sqrt{I_R^2 + (I_L - I_C)^2}$

13. $\tan \phi = \frac{I_L - I_C}{I_R}$

14. $Z = \frac{V}{I}$

15. $A = \frac{\pi d^2}{4}$

16. $R_T = R_1 + R_2 + R_3 + \dots + R_n$

17. $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$

18. $C_T = C_1 + C_2 + C_3 + \dots + C_n$

19. $\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_n}$

20. $L_T = L_1 + L_2 + L_3 + \dots + L_n$

21. $\frac{1}{L_T} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \dots + \frac{1}{L_n}$

22. $Q = C V$

23. $R_t = R_0(1 + \alpha t)$

24. $\frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1}$

25. $I_1 = \left(\frac{R_2}{R_1 + R_2} \right) I_T$

26. $V_C = I X_C$

27. $R = \frac{\rho l}{A}$

28. $\tan \phi = \frac{X_L - X_C}{R}$

29. $E = I^2 R t$

30. $n_s = \frac{f}{p}$

31. $E = V + I r$